

Algorithmic Stock Screener: Data Schema & Sorting Logic

1. System Objective

The objective of this module is to ingest a daily list of the top 100 high-volume stocks and run them through a rigid algorithmic sorting matrix. The system categorizes these stocks into three actionable trading strategies based on Price Shock, Volume Shock, and Absolute Delivery Shock.

2. Required Input (JSON Payload)

For each of the top 100 stocks, the data extraction tool must output a JSON object containing the following schema.

```
{
  "ticker_symbol": "STRING",
  "trade_date": "YYYY-MM-DD",
  "move_category": "GAP_UP | INTRADAY_TREND",
  "today_metrics": {
    "price_change_pct": "DECIMAL",
    "total_volume": "INTEGER",
    "delivery_quantity": "INTEGER",
    "delivery_pct": "DECIMAL"
  },
  "baseline_metrics": {
    "recent_buffer_days_excluded": 5,
    "volume_base_avg": "INTEGER",
    "delivery_qty_base_avg": "INTEGER",
    "highest_high_30d": "DECIMAL"
  },
  "calculated_ratios": {
    "volume_shock_ratio": "DECIMAL",
    "delivery_shock_ratio": "DECIMAL"
  },
  "historical_data": [
    {
      "date": "YYYY-MM-DD",
      "open": "DECIMAL",
      "high": "DECIMAL",
      "low": "DECIMAL",
      "close": "DECIMAL",
      "volume": "INTEGER",
      "delivery_qty": "INTEGER"
    }
  ]
}
```

Developer Note on Baseline & Calculated Ratios:

- **The Buffer Rule:** When calculating `volume_base_avg` and `delivery_qty_base_avg` over the historical window (e.g., 30 days), the system **MUST exclude the most recent N days** (defined by `recent_buffer_days_excluded`, recommended 5-10 days). This prevents a live accumulation phase from poisoning its own baseline.
- $$\text{volume_shock_ratio} = \frac{\text{today_metrics.total_volume}}{\text{baseline_metrics.volume_base_avg}}$$

- `delivery_shock_ratio = today_metrics.delivery_quantity / baseline_metrics.delivery_qty_base_avg`

3. The Classification Matrix (Logic Engine)

The system iterates through the daily JSON array and classifies each stock into ONE of the following four buckets.

Bucket 1: Strategy 1 - The Liquidity Climax (Mean Reversion / Short Play)

Logic: Identifies extreme retail FOMO and HFT churn. Massive volume shock, but weak delivery shock.

- **Condition A (Price):** `price_change_pct > 10.0`
- **Condition B (Volume Shock):** `volume_shock_ratio > 15.0`
- **Condition C (Delivery Shock Gap):** `delivery_shock_ratio < 5.0`
- **Action Flag:** "SHORT_CANDIDATE"

Bucket 2: Strategy 2 - Wyckoff Phase D Breakout (Structural Long)

Logic: Identifies smart money aggressively marking up the price. Absolute Delivery Multiplier paces with the Volume Multiplier.

- **Condition A (Price):** `price_change_pct >= 5.0 AND <= 10.0`
- **Condition B (Volume Shock):** `volume_shock_ratio >= 5.0 AND <= 15.0`
- **Condition C (Absolute Delivery Absorption):** `delivery_shock_ratio >= 5.0`
- **Condition D (Market Structure):** `Today's High > highest_high_30d`
- **Action Flag:** "WYCKOFF_LONG"

Bucket 3: Strategy 3 - Sustainable Momentum (Trend Following Long)

Logic: Identifies quiet, steady institutional accumulation. Elevated but controlled volume, accompanied by a massive surge in absolute delivery compared to the buffered baseline.

- **Condition A (Price):** `price_change_pct >= 2.0 AND <= 6.0`
- **Condition B (Volume Shock):** `volume_shock_ratio >= 2.0 AND <= 5.0`
- **Condition C (Absolute Delivery Absorption):** `delivery_shock_ratio >= 3.0`
- **Action Flag:** "TREND_LONG"

Bucket 4: The Noise (Discard)

Logic: Fails to meet every condition for Bucket 1, 2, or 3.

- **Action Flag:** "DISCARD"